IMPERIAL

Conformal Risk Control for Safety-Critical Wildfire Evacuation Mapping

A Comparative Study of Tabular, Spatial, and Graph-Based Models

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Why This Matters

False Negatives Kill

A fire pixel classified as safe means people stay in a burning zone. Standard ML models miss 48–93% of fires at conventional thresholds.

Shadow Evacuations

Over-broad perimeters cause residents outside the official zone to self-evacuate, congesting roads and trapping those in genuine danger (Zhao et al., 2022).

92.8%

of fires missed by LightGBM at standard threshold ($p \geq 0.5$)

61.0%

missed by U-Net

48.5%

missed by ResGNN

No model is safe at $p \geq 0.5$. We need distribution-free guarantees.

The Research Gap

Wildfire ML

Extensive work on spread prediction (CNNs, GNNs, U-Nets). All evaluate on accuracy metrics: F1, IoU, AUROC.

Conformal Prediction

CRC provides distribution-free safety guarantees. Applied to medical imaging, autonomous driving, NLP.

The Missing Link

The domain where false negatives are most dangerous has no formal safety guarantees.

First application of conformal risk control to wildfire spread prediction

Related Work

Wildfire ML

- Jain & Coogan (2020) — Review
- Radke et al. (2019) — FireCast
- Huot et al. (2022) — NDWS dataset
- **All use accuracy metrics only**

Conformal Methods

- Vovk et al. (2005) — Foundations
- Romano et al. (2020) — Adaptive sets
- Angelopoulos et al. (2022) — CRC
- **Not applied to geospatial/disaster**

Calibration & UQ

- Guo et al. (2017) — Miscalibration
- Platt (1999) — Platt scaling
- No finite-sample guarantees
- **Heuristic uncertainty only**

Safety-Critical ML

- Amodei et al. (2016) — AI safety
- FN cost $\gg$ FP cost in evacuation
- Formal guarantees needed, not just empirical perf.
- **Wildfire: no guarantees until now**

Our Approach

NDWS Dataset

18,545 samples
64×64 patches
12 channels

3 Model Families

LightGBM
Tiny U-Net
ResGNN-UNet

CRC Calibration

FNR ≤ 0.05
Distribution-free
Finite-sample

Safe Decisions

95%+ coverage
Tight zones
3-way triage

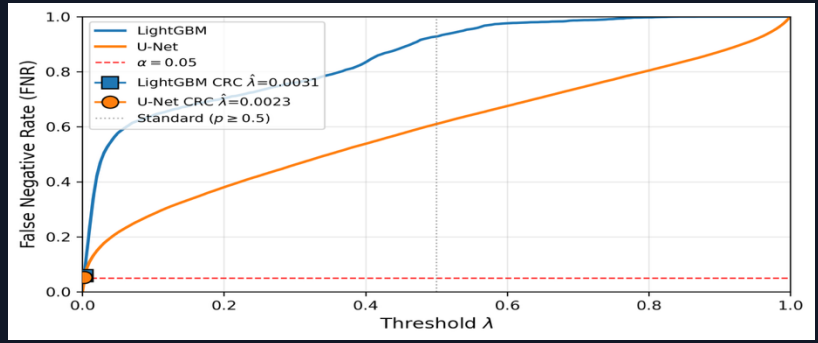
Key Design Decisions

- Deterministic splits (seed 42, 70/15/15) — calibration set never used in training
- Same data + splits across all models — fair architectural comparison
- CRC applied uniformly post-hoc — model-agnostic safety wrapper

How CRC Works

The Intuition

1. Collect predicted probabilities on held-out calibration set
2. Sort the probabilities of true fire pixels
3. Pick the α-quantile as threshold λ̂ (the "safety cutoff") and Apply λ̂ unchanged to test data → guaranteed FNR ≤ α



FNR vs. threshold: U-Net stays safe over a much wider range

The Guarantee

$$E[FNR(\hat{\lambda})] \leq \alpha = 0.05$$

Holds regardless of model quality, calibration, or architecture

Key Insight

CRC decouples safety from efficiency.

Model quality → evacuation zone size

CRC → fire coverage guarantee

Three Model Families

AUROC 0.854

LightGBM

Tabular Baseline

- Pixel-level features (12 channels)
- Each pixel treated independently
- 5% pixel subsampling for memory
- Binary cross-entropy objective

AUROC 0.969

Tiny U-Net

Spatial CNN

- 470K parameters
- 2 encoder + 2 decoder blocks
- Skip connections, 1×1 conv head
- AdamW + cosine annealing, 50 epochs

AUROC 0.951

ResGNN-UNet

Graph + CNN Hybrid

- 229K parameters
- CNN encoder → GAT bottleneck → decoder
- 3 GATConv layers, 4 heads each
- Early-stopped at epoch 12/25

Increasing complexity: pixel-independent → local spatial → graph propagation

Three-Way CRC Extension

SAFE

$\hat{p} < \lambda_{min}$

No action needed

→

MONITOR

$\lambda_{min} \leq \hat{p} < \lambda_{max}$

Human review

→

EVACUATE

$\hat{p} \geq \lambda_{max}$

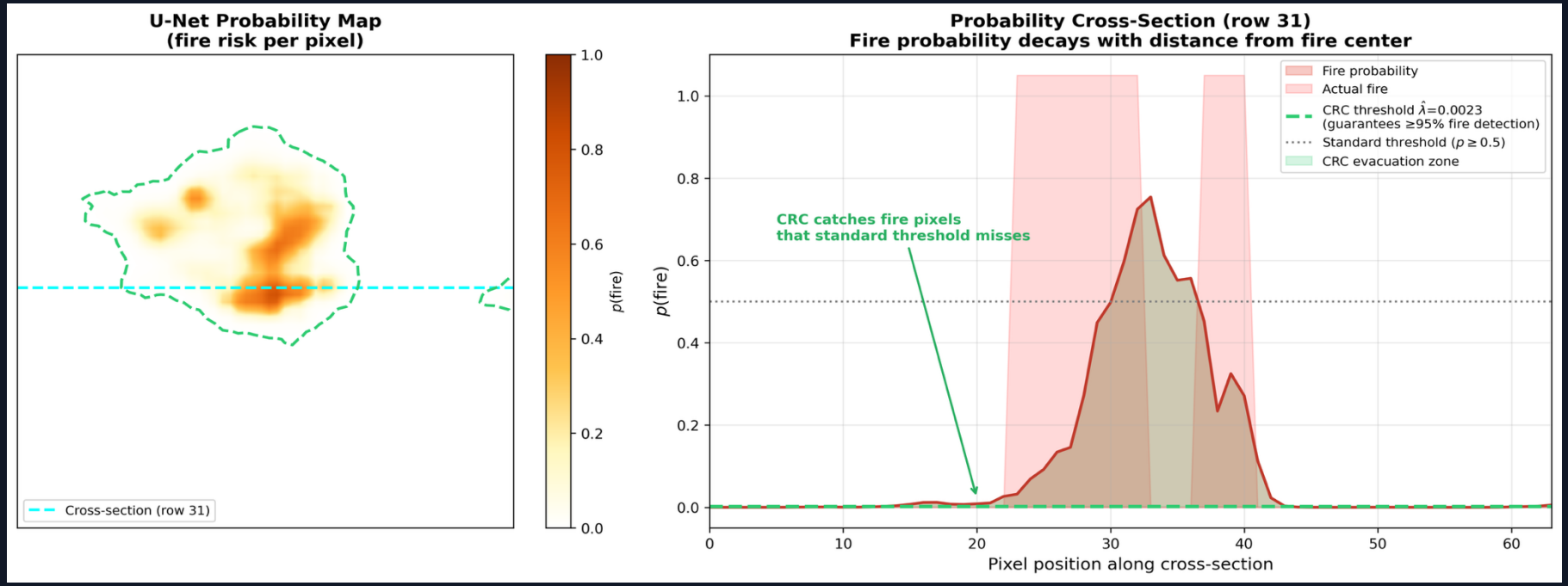
Immediate evacuation

Cost Parameters

- $c_{fn} = 5$ (missed fire cost) vs $c_{fp} = 1$ (false alarm cost)
- Asymmetric: missing fire risks lives; over-alerting causes logistics issues
- Shift interval $\rho \in [0.9, 1.1]$ for mild prevalence variation

Routes uncertain pixels to human review instead of forcing binary automated decisions

Why Standard Thresholds Fail



Probability cross-section through a U-Net prediction. Standard threshold ($p \geq 0.5$, gray) misses most of the fire region. CRC threshold ($\hat{\lambda} = 0.002$, green) captures it entirely.

Results: CRC Delivers Safety

Method	Params	λ̂	Coverage ↑	FNR ↓	Set Size ↓	AUROC ↑
LGBM (p≥0.5)	—	.500	7.2%	.928	0.1%	.854
LGBM + CRC	—	.003	94.1%	.059	62.6%	.854
U-Net (p≥0.5)	470K	.500	39.0%	.610	0.7%	.969
U-Net + CRC	470K	.002	94.7%	.053	14.9%	.969
ResGNN (p≥0.5)	229K	.500	51.5%	.485	—	.951

4.2×

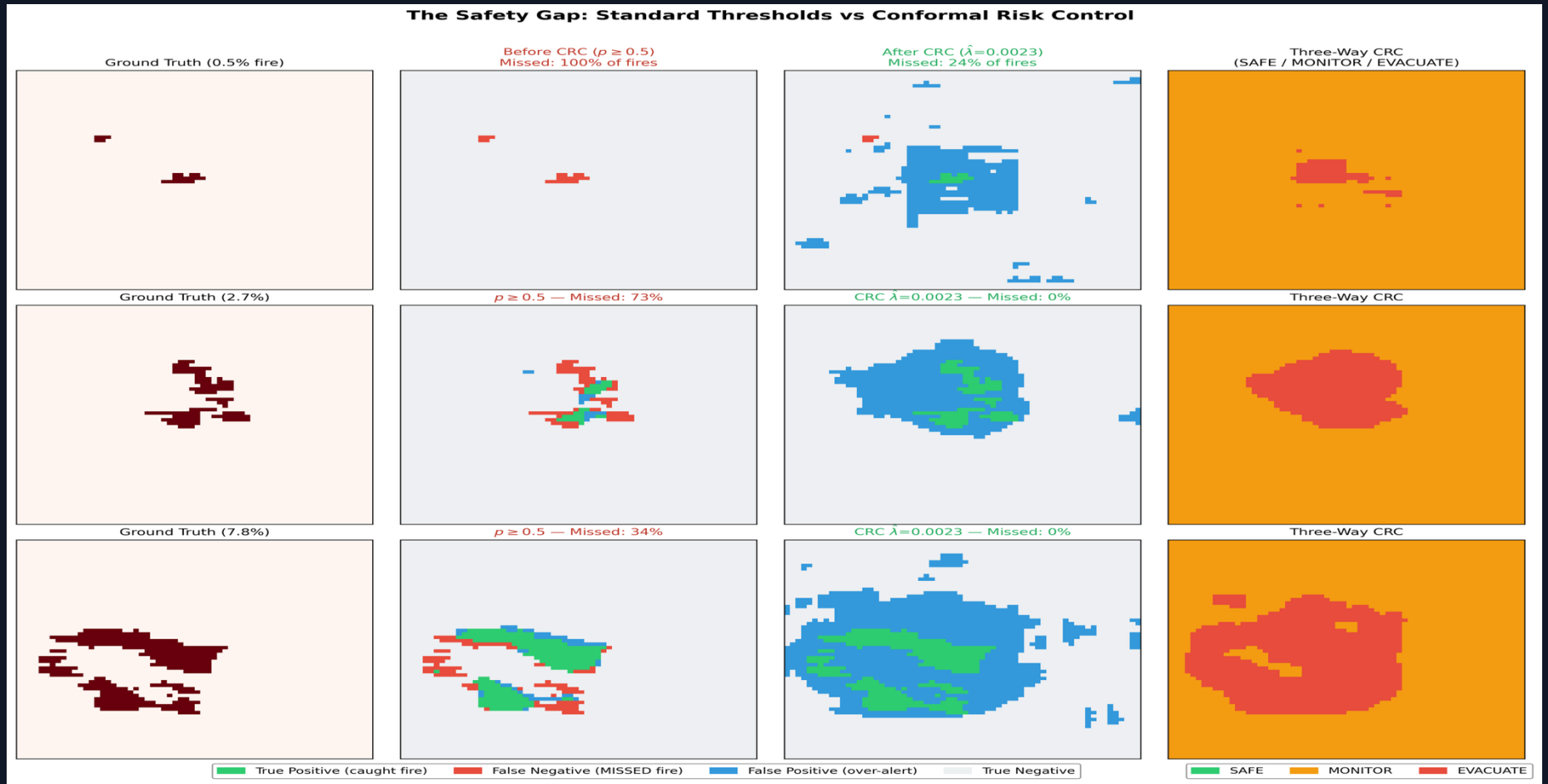
smaller evacuation zones
U-Net vs LightGBM
under same safety guarantee

95% +

fire coverage achieved
by both models with CRC
regardless of architecture

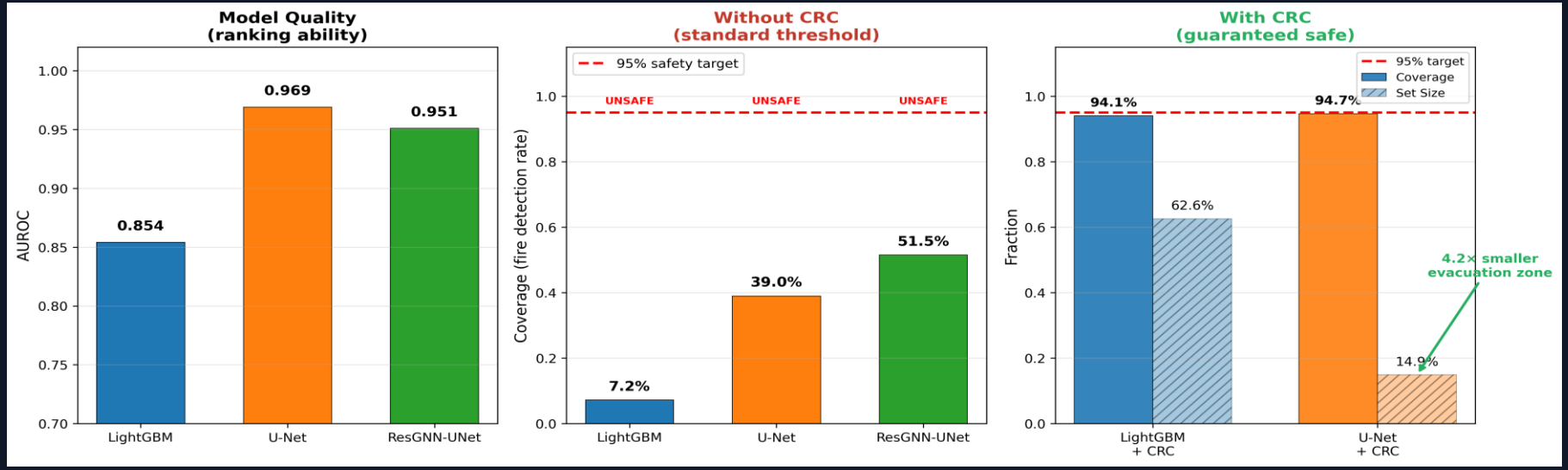
Before vs. After CRC

Standard threshold: 34–100% fires missed → CRC: 0–2% fires missed



The Central Finding

Model architecture determines evacuation efficiency. CRC determines safety.



Complexity ≠ better separation: ResGNN-UNet (229K, graph attention) achieves lower AUROC than Tiny U-Net (470K, standard conv) on 64×64 patches.

Three-Way Zone Analysis

Model	SAFE	MONITOR	EVACUATE
LightGBM	0.0%	95.96%	4.04%
U-Net	0.0%	94.90%	5.10%

The SAFE Zone Collapses

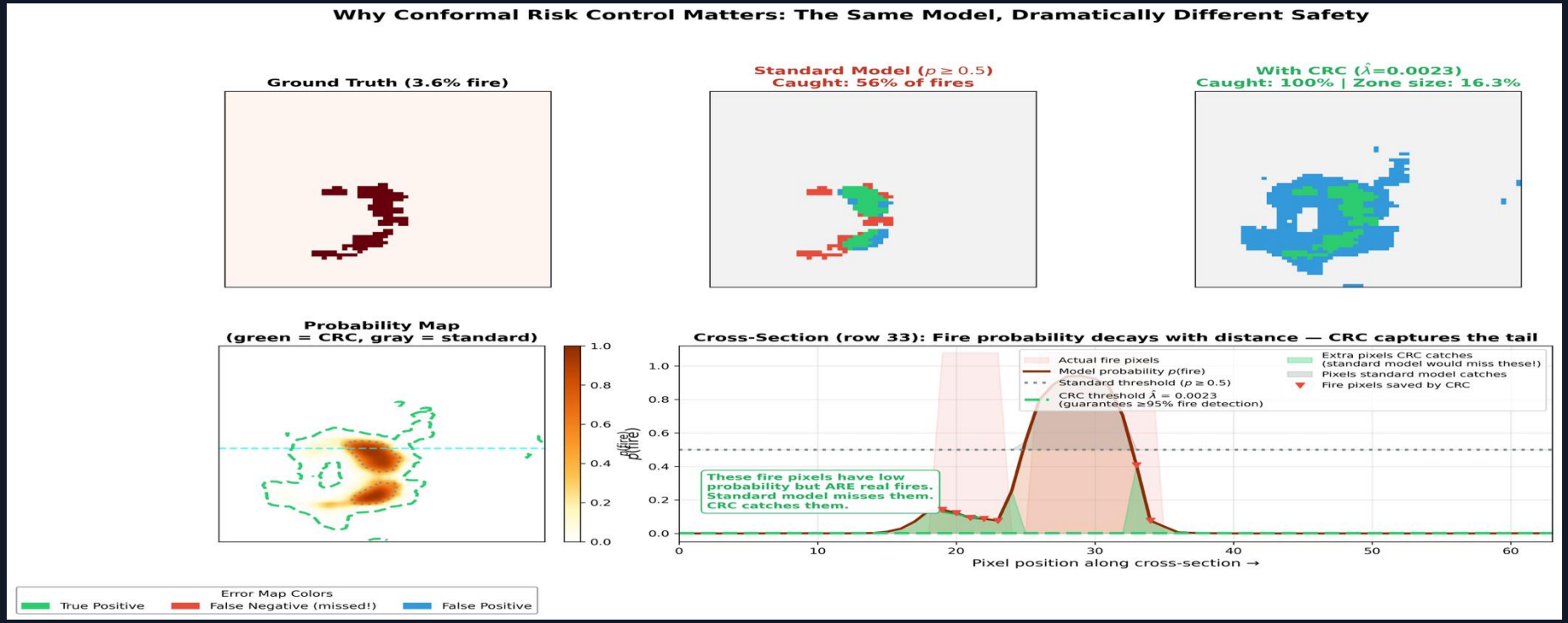
With ~5% fire prevalence:

$$B_pw = \max(5 \times 0.05,\ 1 \times 0.95) = 0.95$$

Dominated by majority-class term $c_fp \cdot \pi_0$. The shift correction exceeds the CRC threshold, pushing λ_min below zero. All non-evacuated pixels routed to human review.

Fundamental limitation of prevalence-weighted bounds in rare-event regimes ($\pi_0 \gg \pi_1$)

CRC Deep Dive: Same Model, Different Thresholds



Standard thresholding (p ≥ 0.5) catches 56% of fires. CRC (λ̂ = 0.002) catches 100%. Green shading marks pixels saved by CRC.

Conclusion

1

No model is safe without CRC

Standard thresholding captures only 7–52% of fires across all architectures.

2

CRC guarantees safety regardless of model

Both calibrated models achieve ≥94% coverage on held-out test data.

3

Model quality determines efficiency, not safety

Tiny U-Net achieves 4.2× tighter evacuation zones than LightGBM under identical CRC guarantee.

4

Complexity without separation is waste

Graph-augmented ResGNN-UNet underperforms standard U-Net in AUROC despite more complex architecture.

A simple model + CRC provides stronger safety AND tighter zones than a complex model evaluated on F1 alone.

Future Work

Class-Conditional CRC

Calibrate each zone boundary independently (FNR-CRC for λ_max, FPR-CRC for λ_min) to resolve SAFE zone collapse under imbalance.

Image-Level CRC

Move from pixel-pooled to per-image FNR losses. Restores theoretical exchangeability, produces operationally meaningful per-image guarantees.

AUROC–Efficiency Formalization

Derive closed-form relationship between model discrimination and evacuation cost. Answer: what AUROC for a target zone size?

Temporal & Geographic Shift

Validate CRC robustness under seasonal/climate shift using temporally stratified splits and multi-region datasets.

Thank You

Questions?

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github.com/baljinnyamday/wildfire-evacuation-crc